PES UNIVERSITY

(Established under Karnataka Act No. 16 of 2013)

100 Feet Ring Road, BSK III Stage, Bengaluru-560 085

UE24MA241 B Linear Algebra and Its Applications
Mini Project rubrics

Create a form to accept their code as submission. PPT not necessary.

Project Demo (5 marks)

- Project Explanation (1 Mark):

Student must clearly explain the working of the code, showing genuine understanding (not just generated content).

- Linear Algebra Concepts (2 Marks):

Correct use and explanation of concepts such as matrix representation, RREF, basis, projections, least squares, eigenvalues, etc.

- Input Explanation (1 Mark):

Input data and its matrix representation must be clearly explained.

- Output Explanation (1 Mark):

Final output should be clearly interpreted with meaning.

Participation Rule:

No Active participation of an individual during demo → -1 mark penalty.

Project Viva (5 marks)

Each student is asked 2 questions

Each question carries 2.5 marks

Total per student = 5 marks

Full Marks (2.5) Concept explained correctly

Partial (1.5 – 2.0) Concept mostly correct

Low (0.5 – 1.0) Basic idea only

Zero (0) No answer / completely incorrect

Sample Viva questions

1. How did you represent your data as a matrix?
2. What do the rows and columns of your matrix represent?
3. Why is matrix representation suitable for your problem?
4. Explain your workflow from input to final output.
5. Which linear algebra concepts did you use?
6. Why did you choose these concepts?
7. What is the purpose of each concept used?
8. What result did each step produce on your data?
9. What is your final output and how should it be interpreted?
10. Did you simplify the matrix? Why was it needed?
11. What is the rank of your matrix and what does it indicate?
12. How do you identify redundancy in your data?
13. What is linear independence and how is it relevant here?
14. What is a basis and why is it important in your project?
15. What is the dimension of your data space?
16. What is orthogonality? Did you use it? Why?
17. What is projection and where did you apply it?
18. What exactly are you projecting and onto what space?
19. How does projection help in solving your problem?
20. What is least squares and why did you use it?
21. What happens if your system has no exact solution?
22. How is least squares related to projection?
23. What are eigenvalues and eigenvectors?
24. How do they help in identifying patterns in your data?
25. What happens if an eigenvalue is zero?
26. Why would you diagonalize a matrix in your project?
27. What assumptions are you making about your data?
28. What happens if your data is noisy or incomplete?
29. How sensitive is your result to small changes in input?
30. What happens if your matrix is singular?
31. Can your project work without one of the steps? Which one and why?
32. Can you replace your method with another concept? What changes?
33. Why is your pipeline arranged in this order?
34. Can any step be skipped or reordered?
35. Which step is most important in your project and why?
36. Which step contributes most to improving the result?

37. What are the limitations of your approach?
38. In which situations will your model fail?
39. How can you improve your project further?
40. Can your method work on a different dataset? Why or why not?